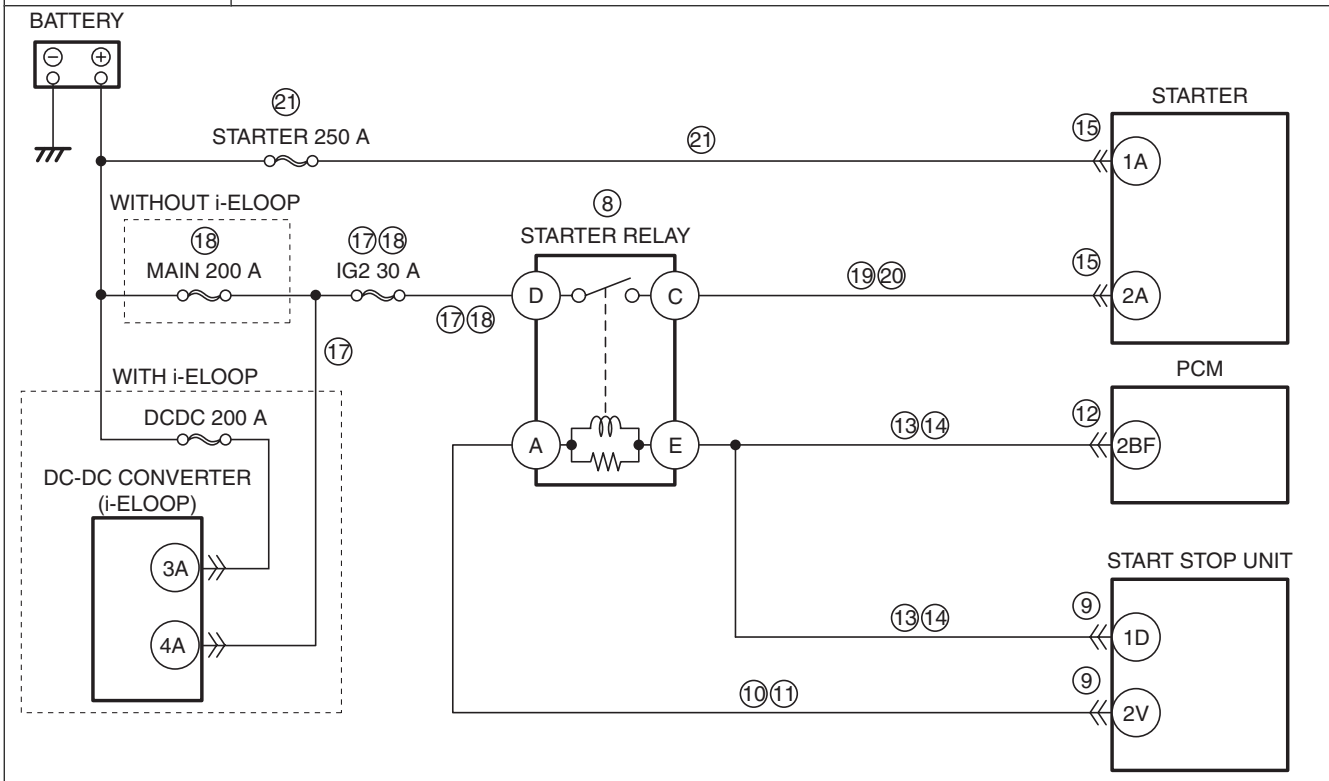
## NO.9 WILL NOT CRANK [SKYACTIV-D 1.5]

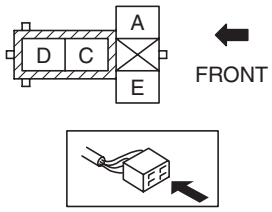
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| 9 | WILL NOT CRANK |
|---|---|
| DESCRIPTION | • Starter does not work. |
| POSSIBLE CAUSE | • Poor connection of push button start connector<br>• Immobilizer system malfunction (short or open circuit, coil antenna malfunction)<br>• Open circuit in wiring harness between the following terminals:<br>— Main relay terminal E—PCM terminal 2K<br>— Main relay terminal C—PCM terminal 2S, 2T, 1DH, 1DL<br>— DLC-2—PCM terminal 2C, 2D<br>• Main relay malfunction (stuck open)<br>• Battery malfunction<br>• Fuse malfunction<br>• Starter relay malfunction<br>• Starter relay related wiring harness malfunction<br>— Between starter relay terminal E and PCM terminal 2BF<br>— Between starter relay terminal E and start stop unit terminal 1D<br>— Between start stop unit terminal 2V and starter relay terminal A<br>• Following circuit malfunction:<br>— Between battery positive terminal and starter terminal 1A<br>— Between battery positive terminal and starter relay terminal D (without i-ELOOP)<br>— Between DC-DC converter (i-ELOOP) terminal 4A and starter relay terminal D (with i-ELOOP)<br>— Between starter relay terminal C and starter terminal 2A<br>• Starter malfunction<br>• Following circuit and/or connector malfunction:<br>— Between push button start terminal A and start stop unit terminal 1AC<br>— Between push button start terminal B and start stop unit terminal 1AE<br>• Push button start malfunction<br>• Mechanical (engine) malfunction<br>— Large mechanical resistance (such as A/C compressor)<br>• PCM malfunction<br><br>**Warning**<br>**• The following troubleshooting flow chart contains the fuel system diagnosis and repair procedures. Read the following warnings before performing the fuel system services:**<br>**— Always keep sparks and flames away from fuel. Fuel can be easily ignited which could cause serious injury or death, and damage to equipment.**<br>**— Fuel line spills and leakage from the pressurized fuel system are dangerous. Fuel can ignite and cause serious injury or death, and damage to property and facilities. Fuel can also irritate skin and eyes. To prevent this, always complete the "Fuel Line Safety Procedure", while referring to the "BEFORE SERVICE PRECAUTION". (See BEFORE SERVICE PRECAUTION [SKYACTIV-D 1.5].)**<br>**— Fuel is highly flammable and dangerous. Fuel line spills and leakage can cause serious injury or death, and damage to equipment. When installing the fuel hose, always refer to the "AFTER SERVICE PRECAUTION" and perform the "Fuel Hose Installation Procedure". (See AFTER SERVICE PRECAUTION [SKYACTIV-D 1.5].)** |

| 9 | WILL NOT CRANK |
|---|---|



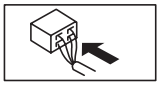

STARTER RELAY



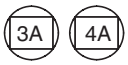


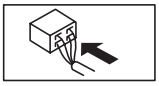

STARTER
WIRING HARNESS-SIDE CONNECTOR



DC-DC CONVERTER (i-ELOOP)
WIRING HARNESS-SIDE CONNECTOR

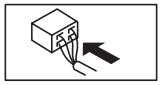


START STOP UNIT
WIRING HARNESS-SIDE CONNECTOR

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1AE | 1AC | 1AA | 1Y | 1W | 1U | 1S | 1Q | 1O | 1M | 1K | 1I | 1G | 1E | 1C | 1A |
| 1AF | 1AD | 1AB | 1Z | 1X | 1V | 1T | 1R | 1P | 1N | 1L | 1J | 1H | 1F | 1D | 1B |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2W | 2U | 2S | 2Q | 2O | 2M | 2K | 2I | 2G | 2E | 2C | 2A |
| 2X | 2V | 2T | 2R | 2P | 2N | 2L | 2J | 2H | 2F | 2D | 2B |

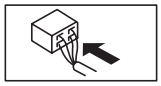


PCM
WIRING HARNESS-SIDE CONNECTOR

| | | | | |
|---|---|---|---|---|
| 2BE | 2AZ | 2AU | 2AP | 2AK |
| 2BF | 2BA | 2AV | 2AQ | 2AL |
| 2BG | 2BB | 2AW | 2AR | 2AM |
| 2BH | 2BC | 2AX | 2AS | 2AN |
| | 2BD | 2AY | 2AT | 2AO |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 2AE | 2AA | 2W | 2S | 2O | 2K | 2G | 2C |
| 2AF | 2AB | 2X | 2T | 2P | 2L | 2H | 2D |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 2AI | 2AG | 2AC | 2Y | 2U | 2Q | 2M | 2I | 2E | 2A |
| 2AJ | 2AH | 2AD | 2Z | 2V | 2R | 2N | 2J | 2F | 2B |



### Diagnostic Procedure

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 1 | **DETERMINE IF MALFUNCTION CAUSE IS IMMOBILIZER SYSTEM OR OTHER**<br>• Are any of the following conditions present?<br>— Engine does not start completely.<br>— PCM DTC P1260:00 is displayed. | Yes | Both conditions present:<br>• Go to Step 3. |
| | | No | Either or other condition present:<br>• Go to the next step. |
| 2 | **INSPECT PUSH BUTTON START CONNECTOR CONNECTION**<br>• Inspect the connection of push button start connector.<br>• Is the push button start connector securely connected to the coil antenna? | Yes | Go to the next step. |
| | | No | Reconnect the push button start securely, then repeat from Step 1. |
| 3 | **VERIFY IMMOBILIZER SYSTEM DTC**<br>• Retrieve the immobilizer system DTCs using the M-MDS.<br>(See DTC INSPECTION [IMMOBILIZER SYSTEM].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [IMMOBILIZER SYSTEM].) |
| | | No | Go to the next step. |
| 4 | **DETERMINE IF MALFUNCTION CAUSE IS i-stop SYSTEM OR OTHER**<br>• Verify the symptom.<br>• Does the engine not restart while the i-stop function is operating? | Yes | Perform the symptom troubleshooting "NO.6 ENGINE DOES NOT RESTART".<br>(See NO.6 ENGINE DOES NOT RESTART [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |
| 5 | **VERIFY THAT COMMUNICATION ERROR MESSAGE IS DISPLAYED**<br>• Retrieve any DTCs using the M-MDS.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Is the communication error message displayed? | Yes | Inspect the following:<br>• Open circuit in wiring harness between main relay terminal E and PCM terminal 2K<br>• Open circuit in wiring harness between main relay terminal C and PCM terminal 2S, 2T, 1DH, 1DL<br>• Main relay (stuck open)<br>• Open or short circuit in wiring harness between DLC-2 and PCM terminal 2C, 2D<br>Repair or replace the malfunctioning part according to the inspection results, then go to Step 26. |
| | | No | Go to the next step. |
| 6 | **INSPECT POWER SUPPLY**<br>• Access the VPWR PID using the M-MDS.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Is the VPWR PID value **B+**? | Yes | Go to the next step. |
| | | No | Inspect the following:<br>• Battery connection<br>• Battery condition<br>(See BATTERY INSPECTION.)<br>• Fuse<br>(See NO.1 BLOWN FUSES [SKYACTIV-D 1.5].)<br>— If there is any malfunction:<br>• Repair or replace the malfunctioning part according to the inspection results, then repeat this step. |
| 7 | **DETERMINE IF MALFUNCTION CAUSE IS STARTER RELAY CONTROL SIGNAL CIRCUIT OR OTHER**<br>• Crank the engine.<br>• Is a clicking sound heard from the starter relay? | Yes | Go to Step 15. |
| | | No | Go to the next step. |
| 8 | **INSPECT STARTER RELAY**<br>• Switch the ignition off.<br>• Remove the starter relay.<br>• Inspect the starter relay.<br>(See RELAY INSPECTION.)<br>• Is the starter relay normal? | Yes | Go to the next step. |
| | | No | Replace the starter relay, then go to Step 26. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 9 | **INSPECT START STOP UNIT CONNECTOR CONDITION**<br>• Disconnect the start stop unit connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is the connector normal? | Yes | Go to the next step. |
| | | No | Repair or replace the connector and/or terminals, then go to Step 26. |
| 10 | **INSPECT FOR SHORT TO GROUND IN PRIMARY POWER SUPPLY OF STARTER RELAY**<br>• Verify that the starter relay is removed.<br>• Verify that the start stop unit connector is disconnected.<br>• Inspect for continuity between starter relay terminal A (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector between start stop unit terminal 2V and starter relay terminal A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>Go to Step 26. |
| | | No | Go to the next step. |
| 11 | **INSPECT FOR OPEN CIRCUIT IN PRIMARY POWER SUPPLY OF STARTER RELAY**<br>• Verify that the starter relay is removed.<br>• Verify that the start stop unit connector is disconnected.<br>• Inspect for continuity between start stop unit terminal 2V (wiring harness-side) and starter relay terminal A (wiring harness-side).<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between start stop unit terminal 2V and starter relay terminal A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 26. |
| 12 | **INSPECT PCM CONNECTOR CONDITION**<br>• Disconnect the PCM connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is the connector normal? | Yes | Go to the next step. |
| | | No | Repair or replace the connector and/or terminals, then go to Step 26. |
| 13 | **INSPECT STARTER RELAY CONTROL CIRCUIT FOR SHORT TO GROUND**<br>• Verify that the starter relay is removed.<br>• Verify that the start stop unit and PCM connectors are disconnected.<br>• Inspect for continuity between starter relay terminal E (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>• Starter relay terminal E—PCM terminal 2BF<br>• Starter relay terminal E—Start stop unit terminal 1D<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>Go to Step 26. |
| | | No | Go to the next step. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 14 | **INSPECT STARTER RELAY CONTROL CIRCUIT FOR OPEN CIRCUIT**<br>• Verify that the starter relay is removed.<br>• Verify that the start stop unit and PCM connectors are disconnected.<br>• Inspect for continuity between the following terminals (wiring harness-side):<br>— Starter relay terminal E—PCM terminal 2BF<br>— Starter relay terminal E—Start stop unit terminal 1D<br>• Is there continuity? | Yes | Inspect the start stop unit.<br>(See START STOP UNIT INSPECTION.)<br>• If there is any malfunction:<br>— Replace the start stop unit, then go to Step 26.<br>(See START STOP UNIT REMOVAL/INSTALLATION.)<br>• If there is no malfunction:<br>— Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>• Starter relay terminal E—PCM terminal 2BF<br>• Starter relay terminal E—Start stop unit terminal 1D<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 26. |
| 15 | **INSPECT STARTER CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the starter connector.<br>• Inspect for poor connection (such as damaged/pulled-out pins, corrosion).<br>• Is the connector normal? | Yes | Go to the next step. |
| | | No | Repair or replace the connector and/or terminals, then go to Step 26. |
| 16 | **DETERMINE IF MALFUNCTION CAUSE IS STARTER OR OTHER**<br>• Verify that the starter connector is disconnected.<br>• Crank the engine.<br>• Measure the voltage at the starter terminal 2A (wiring harness-side)<br>• Is the voltage **B+**? | Yes | Go to Step 21. |
| | | No | **With i-ELOOP:**<br>• Go to the next step.<br>**Without i-ELOOP:**<br>• Go to Step 18. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 17 | **INSPECT FOR SHORT TO GROUND AND OPEN CIRCUIT IN SECONDARY (STARTER POWER SUPPLY) OF STARTER RELAY**<br>• Switch the ignition off.<br>• Remove the starter relay.<br>• Verify that the starter connector is disconnected.<br>• Measure the voltage at the starter relay terminal D (wiring harness-side) and body ground.<br>• Is the voltage B+? | Yes | Go to the next step. |
| | | No | Remove the service plug.<br>(See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)<br>Inspect the IG2 30 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and starter relay terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>• Replace the malfunctioning fuse.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>• Replace the malfunctioning fuse.<br>• If the fuse is damaged:<br>— Replace the malfunctioning fuse.<br>• If all fuses are normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and starter relay terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Install the service plug.<br>(See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)<br>Go to Step 26. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 18 | **INSPECT FOR SHORT TO GROUND AND OPEN CIRCUIT IN SECONDARY (STARTER POWER SUPPLY) OF STARTER RELAY**<br>• Switch the ignition off.<br>• Remove the starter relay.<br>• Verify that the starter connector is disconnected.<br>• Measure the voltage at the starter relay terminal D (wiring harness-side)<br>• Is the voltage **B+**? | Yes | Go to the next step. |
| | | No | Inspect the MAIN 200 A fuse and IG2 30 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and starter relay terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>• Replace the fuse.<br>• If the fuse is damaged:<br>— Replace the fuse.<br>• If all fuses are normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and starter relay terminal D.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 26. |
| 19 | **INSPECT STARTER POWER SUPPLY CIRCUIT FOR SHORT TO GROUND**<br>• Verify that the starter relay is removed.<br>• Verify that the starter connector is disconnected.<br>• Inspect for continuity between starter relay terminal C (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector between starter relay terminal C and starter terminal 2A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>Go to Step 26. |
| | | No | Go to the next step. |
| 20 | **INSPECT STARTER POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT**<br>• Verify that the starter relay is removed.<br>• Verify that the starter connector is disconnected.<br>• Inspect for continuity between starter relay terminal C (wiring harness-side) and starter terminal 2A (wiring harness-side).<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between starter relay terminal C and starter terminal 2A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 26. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 21 | **INSPECT STARTER POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT**<br>• Verify that the starter connector is disconnected.<br>• Switch the ignition off.<br>• Measure the voltage at the starter terminal 1A (wiring harness-side)<br>• Is the voltage **B+**? | Yes | Go to the next step. |
| | | No | Inspect the STARTER 250 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between STARTER 450 A fuse and starter terminal 1A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>• Replace the fuse.<br>• If the fuse is damaged:<br>— Replace the fuse.<br>• If the fuse is normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and starter terminal 1A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 26. |
| 22 | **INSPECT STARTER**<br>• Inspect the starter. (See STARTER INSPECTION [SKYACTIV-D 1.5].)<br>• Is the starter normal? | Yes | Go to the next step. |
| | | No | Replace the starter, then go to Step 26. (See STARTER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) |
| 23 | **INSPECT IMMOBILIZER SYSTEM RELATED CIRCUIT**<br>• Inspect the following wiring harness and connectors:<br>— Between push button start terminal A and start stop unit terminal 1AC<br>— Between push button start terminal B and start stop unit terminal 1AE<br>• Are the wiring harness and connectors normal? | Yes | Go to the next step. |
| | | No | Repair or replace the malfunctioning part according to the inspection results, then go to Step 26. |
| 24 | **DETERMINE IF MALFUNCTION IS DUE TO EXCESSIVE ENGINE SPEED RESISTANCE**<br>• Rotate the crankshaft pulley lock bolt clockwise using a wrench. (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-D 1.5].)<br>• Can bolts be rotated? | Yes | Go to Step 26. |
| | | No | Go to the next step. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 25 | **INSPECT FOR MALFUNCTION DUE TO EXCESSIVE MECHANICAL RESISTANCE OF ENGINE ACCESSORIES**<br>• Remove all drive belts from engine accessories. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br><br>**Caution**<br>**• Do not run the engine for long periods with the drive belts of engine accessories removed. Otherwise the engine could be damaged from overheating.**<br>• Start the engine.<br>• Is cranking possible? (Does the engine start?) | Yes | Repair or replace the malfunctioning part according to the inspection results, then go to the next step. (Large mechanical resistance in engine accessories such as the A/C compressor.) |
| | | No | Go to the next step. |
| 26 | Verify the test results.<br>• If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].)<br>• If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis.<br>— If the vehicle is repaired, troubleshooting is completed.<br>— If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) | | |